



Started on	Thursday, 8 October 2026, 9:35 AM
State	Finished
Completed on	Thursday, 8 October 2026, 9:37 AM
Time taken	2 mins 22 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1

Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5	1
1 1 2 3 4	

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2  int main() {
3      int n;
4      scanf("%d", &n);
5      int a[n];
6      for (int i = 0; i < n; i++) {
7          scanf("%d", &a[i]);
8      }
9      for (int i = 0; i < n; i++) {
10         for (int j = i + 1; j < n; j++) {
11             if (a[i] == a[j]) {
12                 printf("%d", a[i]);
13                 return 0;
14             }
15         }
16     }
17
18     return 0;
19 }
20
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Thursday, 8 October 2026, 9:39 AM
State	Finished
Completed on	Thursday, 8 October 2026, 9:39 AM
Time taken	25 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1

Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int a[n];
8
9      for (int i = 0; i < n; i++)
10         scanf("%d", &a[i]);
11
12     int slow = a[0];
13     int fast = a[0];
14
15     do {
16         slow = a[slow];
17         fast = a[a[fast]];
18     } while (slow != fast);
19
20     slow = a[0];
21
22     while (slow != fast) {
23         slow = a[slow];
24         fast = a[fast];
25     }
26
27     printf("%d", slow);
28
29     return 0;
30 }
31
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓

	Input	Expected	Got	
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Thursday, 8 October 2026, 7:35 PM
State	Finished
Completed on	Thursday, 8 October 2026, 7:36 PM
Time taken	45 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  int main() {
3      int T;
4      scanf("%d", &T);
5      while (T--) {
6          int m, n;
7          scanf("%d", &m);
8          int a[m];
9          for (int i = 0; i < m; i++)
10             scanf("%d", &a[i]);
11         scanf("%d", &n);
12         int b[n];
13         for (int i = 0; i < n; i++)
14             scanf("%d", &b[i]);

```



```
15 |         for (int i = 0; i < m; i++) {
16 |             for (int j = 0; j < n; j++) {
17 |                 if (a[i] == b[j]) {
18 |                     printf("%d ", a[i]);
19 |                     break;
20 |                 }
21 |             }
22 |         }
23 |
24 |         printf("\n");
25 |     }
26 |     return 0;
27 | }
28 |
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Thursday, 8 October 2026, 7:36 PM
State	Finished
Completed on	Thursday, 8 October 2026, 7:40 PM
Time taken	4 mins 24 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2 int main() {
3     int T;
4     scanf("%d", &T);
5     while (T--) {
6         int m, n;
7         scanf("%d", &m);
8         int a[m];
9         for (int i = 0; i < m; i++)
10             scanf("%d", &a[i]);
11         scanf("%d", &n);
12         int b[n];
13         for (int i = 0; i < n; i++)
14             scanf("%d", &b[i]);

```

```
15     int i = 0, j = 0;
16     while (i < m && j < n) {
17         if (a[i] == b[j]) {
18             printf("%d ", a[i]);
19             i++;
20             j++;
21         }
22         else if (a[i] < b[j]) {
23             i++;
24         }
25         else {
26             j++;
27         }
28     }
29     printf("\n");
30 }
31 return 0;
32 }
33 }
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Thursday, 8 October 2026, 7:41 PM
State	Finished
Completed on	Thursday, 8 October 2026, 7:43 PM
Time taken	2 mins 42 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  int main() {
3      int n, k;
4      scanf("%d", &n);
5      int a[n];
6      for (int i = 0; i < n; i++)
7          scanf("%d", &a[i]);
8      scanf("%d", &k);
9      for (int i = 0; i < n; i++) {
10         for (int j = 0; j < n; j++) {
11             if (i != j && a[j] - a[i] == k) {
12                 printf("1");
13                 return 0;
14             }
15         }
16     }
17     printf("0");
18     return 0;
19 }
20

```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Thursday, 8 October 2026, 7:44 PM
State	Finished
Completed on	Thursday, 8 October 2026, 7:49 PM
Time taken	5 mins 13 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  int main() {
3      int n, k;
4      scanf("%d", &n);
5      int a[n];
6      for (int i = 0; i < n; i++)
7          scanf("%d", &a[i]);
8      scanf("%d", &k);
9      for (int i = 0; i < n; i++) {
10         for (int j = 0; j < n; j++) {
11             if (i != j && a[j] - a[i] == k) {
12                 printf("1");
13                 return 0;
14             }
15         }
16     }
17     printf("0");
18     return 0;
19 }
20

```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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